PharmAchieve

INSOMNIA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION…

Providing Care: Clinical Care	Providing Care: Distribution	Knowledge & Expertise
Communication & Collaboration	Leadership & Stewardship	Professionalism

LEGEND

5-HT	5-Hydroxytryptamine (Serotonin)
ADHD	Attention-Deficit/Hyperactivity Disorder
AE	Adverse Effect
BPH	Benign Prostate Hypertrophy
CBT	Cognitive Behavioral Therapy
CBT-I	Cognitive Behavioral Therapy for Insomnia
CDSA	Controlled Drug and Substance Act
C/I	Contraindication
CNS	Central Nervous System
CVD	Cardiovascular Disease
DORA	Dual Orexin Receptor Antagonist
EEG	Electroencephalogram
GABA	Gamma-aminobutyric Acid
GERD	Gastroesophageal Reflux Disease
GI	Gastrointestinal
H1	Histamine-1
IOP	Intraocular Pressure
MOA	Mechanism of Action
MOAI	Monoamine Oxidase Inhibitors
NREM	Non-Rapid Eye Movement
ODT	Orally Disintegrating Tablet
OTC	Over-The-Counter
PK	Pharmacokinetic
PO	Per Os (Oral)
REM	Rapid Eye Movement
SSRI	Selective Serotonin Reuptake Inhibitor
SNRI	Selective Norepinephrine Reuptake Inhibitor
TCA	Tricyclic Antidepressant

PATHOPHYSIOLOGY

FYI

Circadian rhythms influence sleep patterns

THERE ARE TWO TYPES OF SLEEP:

NREM SLEEP (non-rapid eye movement)	REM SLEEP (rapid eye movement)
• Normally makes up 75% of total adult sleep time • Consists of 3 stages with each stage taking you deeper into sleep • Brain waves progressively slow down • Commonly occurs within the first hour of sleep • Dreaming is rare	• Normally makes up 20-25% of total adult sleep time • Eyes move rapidly and limbs are temporarily paralyzed • Associated with dreaming • Commonly occurs within the second hour of sleep

Melatonin is the chief chemical/hormonal mediator of the sleep cycle

DIAGNOSIS OF INSOMNIA

- Insomnia is the dissatisfaction with sleep quality or quantity that is associated with 1 or more of the following features: difficulty falling asleep, difficulty staying asleep or early morning awakening without being able to return to sleep
- Occurs at least 3 nights per week for at least 3 months
- Causes significant distress or functional impairment (e.g. social, occupational, educational…etc.)
- Not related to substance use and can not be better explained by co-existing medical or psychiatric conditions
    - Presence of comorbidities should be screened and treated along with insomnia

ACUTE/SHORT-TERM INSOMNIA	CHRONIC INSOMNIA
< 3 months and usually due to a stressful event, environmental disturbance or disruption of the sleep/wake cycle	≥ 3 months with symptoms associated with difficulty falling asleep, staying asleep and/or compromised daytime function

RISK FACTORS AND CAUSES

Risk factors

- Increased age
- Females
- Emotional or physical distress
- Long distance, high-speed travel
- Medical conditions or drugs

Medical conditions causing insomnia

- Psychiatric disorders
- Cardiovascular disorders
- GI disorders
- Cancer
- Chronic pain
- Hormonal disorders
- Neurologic disorders

Drug causes

- CNS Stimulants
- Antidepressants (SSRI/SNRI)
- Opioids in combination with caffeine
- Corticosteroids (PO or Inhaled)
- Decongestants
- Hormonal therapy
- Drugs affecting the adrenergic, cholinergic, dopaminergic pathways
- Other: Alcohol, Caffeine, Cannabis, Nicotine

MANAGEMENT OF INSOMNIA: GOALS OF THERAPY

- ✓ Reduce daytime impairment
- ✓ Restore normal sleep pattern and improve daytime functioning
- ✓ Resolve contributing factors to insomnia
- ✓ Prevent chronic insomnia
- ✓ Minimize side effects of pharmacotherapy

NON-PHARMACOLOGICAL MEASURES

- CBT-I is recommended as first-line therapy for chronic insomnia, and its 5 components include:

Sleep Hygiene

- Limit nicotine, alcohol and caffeine
- Engage in regular exercise (avoid in late evening)
- Avoid large meals before bed
- Avoid bright lights or light-emitting devices hours before bedtime

Sleep Restriction

- Do not go to bed unless tired
- Avoid daytime naps
- Maintain a consistent sleep schedule

Stimulus Control

- Limit bedroom use to sleep and intimacy (no electronic devices, eating, etc.)
- If unable to sleep, do not stay in bed

Cognitive Therapy

- Set reasonable sleep expectations
- Address negative thoughts and emotions contributing to insomnia

Relaxation Therapy

- Attempt breathing exercises, meditation, and progressive relaxation prior to bed

MANAGEMENT OF ACUTE INSOMNIA

- Usually results from psychological or physiological stress. Patients are often able to identify the precipitating cause
- Often self-limited

Approach to treatment:

- Assess level of distress
- Identify the stressor(s) and discuss role in sleep disruption
- Reduce risk of progressing to chronic insomnia
- Nonpharmacological therapy is aimed at reducing psychological and physical stress contributing to sleeplessness
- Short-term pharmacotherapy is appropriate for severe insomnia to minimize daytime complications
- Follow up in 2 to 4 weeks to reassess sleep-related symptoms and reinforce non-pharmacological therapy

PHARMACOLOGICAL MEASURES: GENERAL PRINCIPLES

❑ Pharmacotherapy is considered adjunctive therapy to CBT-I

- Behavioural management remains the focus of treatment
- Drug therapy alone is not recommended as the sole management for insomnia
- Initiate therapy at lowest effective dose for the shortest duration (e.g. 1-2 weeks)
- Long-term use of benzodiazepines and Z-drugs (e.g. > 4 weeks) is not recommended
- Avoid benzodiazepines in elderly patients due to adverse outcomes like falls, fractures and side effects
- In special cases (e.g. refractory or severe insomnia resistant to CBT-I), long-term use of therapy may be appropriate but requires regular follow-up

PHARMACOLOGICAL TREATMENT OPTIONS

- Non-prescription/OTC drugs
- Benzodiazepines
- Z-drugs
- Trazodone and Doxepin
- Dual-orexin antagonists

PHARMACOLOGICAL OPTIONS - PRESCRIPTIONS

Health-Canada Approved Agents	Off-Label - Evidence lacking, not recommended (sometimes used in the presence of co-morbidities)
Benzodiazepines: • Temazepam, flurazepam, nitrazepam, triazolam Benzodiazepine Receptor Agonists (Z-drugs): • Eszopiclone (Lunesta®), zolpidem (Sublinox®), zopiclone (Imovane®) Dual Orexin Receptor Antagonists (DORA): • Lemborexant (Dayvigo®), daridorexant (Quviviq®) Tricyclic Antidepressants (TCAs): • Doxepin (Silenor®) at low doses (3-6 mg)	Sedating antidepressants: • Mirtazapine • Trazodone Second-generation anti-psychotics: • e.g. Quetiapine Other benzodiazepines: • Diazepam • Clonazepam • Lorazepam

PHARMACOLOGICAL MEASURES: NON-PRESCRIPTION

1st generation antihistamines	Melatonin	Others
• E.g. diphenhydramine, dimenhydrinate, doxylamine, chlorpheniramine • Not intended for long-term use, tolerance may develop with extended use • AEs: anticholinergic effects (sedation, dry mouth, blurred vision, urinary retention, constipation) increased intraocular pressure, morning drowsiness and grogginess, confusion • C/I: increased IOP, delirium, urinary obstruction	• Variable, insufficient evidence • Modest effect on ↓ sleep onset latency, ↑ total sleep time and overall sleep quality • Used for jetlag • AEs: fatigue, headache, nausea, dizziness, bradycardia (high dose) • No physical tolerance and dependence • Not recommended during pregnancy and breastfeeding	• E.g. valerian root/extract, chamomile, L-tryptophan • Insufficient evidence to support efficacy • AEs: dizziness, GI upset, headache • Valerian has been associated with hepatoxicity cases • Valerian is not recommended during pregnancy/breastfeeding

BENZODIAZEPINES

BZD indicated for Insomnia in Canada: flurazepam, temazepam, nitrazepam and triazolam

- MOA: Allosteric modulators at the GABA-A receptor complex to enhance the normal inhibitory action of GABA
- For short-term (< 4 weeks), episodic insomnia unresponsive to lifestyle changes and sleep hygiene
- AEs: tolerance, dependence, withdrawal, fall risk, memory performance and cognitive impairment, delirium, respiratory depression and rebound insomnia
- Intermediate-acting agents are preferred: Temazepam has a low to moderate risk of morning hangover. AEs include dizziness, memory impairment, falls, and confusion
- AVOID:
  - Nitrazepam and flurazepam have longer half-lives, thus dosing may lead to accumulation and hangover effects (avoid in the elderly)
  - Triazolam is short-acting with benefits for sleep-onset insomnia but has a higher risk of abuse, dependence and rebound insomnia. It can also cause amnesia
  - BZD in the elderly, if possible, due to risk of AEs like falls, fractures and cognitive effects

BENZODIAZEPINE RECEPTOR AGONISTS (Z-DRUGS) - OVERVIEW

- Zopiclone, zolpidem and eszopiclone are available in Canada. All carry Health-Canada approval for insomnia
- MOA: Agonist at benzodiazepine receptor with greater selectivity for certain GABA-A subunit subtypes
- Similar therapeutic effects on overall sleep as benzodiazepines, but fewer adverse effects of cognitive impairment, dependence, tolerance and rebound insomnia
- For short-term insomnia unresponsive to non-pharmacological interventions
- General adverse effects: bitter/metallic taste, severe drowsiness, impaired coordination
- Warnings: complex sleep behaviours (e.g. night eating, sleep-walking with no recollection), dependence, behavioural change, amnesia, withdrawal, concomitant use with opioids and alcohol
- C/I: comorbid myasthenia gravis, severe hepatic insufficiency, severe respiratory impairment

BENZODIAZEPINE RECEPTOR AGONISTS (Z-DRUGS)

Drugs	Clinical Pearls
Zopiclone (Imovane®)	• Wait at least 12 hours after administration before driving or engaging in tasks requiring full mental capacity • Not recommended in pregnancy, breastfeeding
Eszopiclone (Lunesta®)	• S-isomer of zopiclone • More potent, quicker onset, longer duration of action, and less hangover effects than zopiclone • Decreased absorption if taken with heavy, high-fat meal • Avoid concomitant use of potent CYP3A4 inhibitors • Wait at least 12 hours after administration before driving or engaging in tasks requiring full mental capacity • Not recommended in breastfeeding, limited pregnancy data, use if benefits > risk
Zolpidem (Sublinox®) Sublingual ODT	• CDSA schedule IV • Wait at least 8 hours after administration before driving or engaging in tasks requiring full mental capacity • Place under tongue to allow disintegration • Decreased absorption if taken with food • Avoid in pregnancy, not recommended in breastfeeding

BINDING AFFINITY OF Z-DRUGS AND BENZODIAZEPINES

FYI

<table>
<tr><th colspan="2">Distribution and Functional Effect of GABA-A Receptor Alpha Subunits</th><th colspan="4">Affinity for Receptor</th></tr>
<tr><th>Subunit</th><th>Role</th><th>Benzodiazepine</th><th>Zopiclone</th><th>Eszopiclone</th><th>Zolpidem</th></tr>
<tr><td>Alpha 1</td><td>Hypnotic, anticonvulsant, memory</td><td rowspan="4">High</td><td>Moderate-high</td><td>Moderate-High</td><td>High</td></tr>
<tr><td>Alpha 2</td><td>EEG activity, anxiolytic</td><td>Moderate</td><td rowspan="2">High</td><td rowspan="2">Low-Moderate</td></tr>
<tr><td>Alpha 3</td><td>Sleep, anxiolytic, antidepressant</td><td>Low-moderate</td></tr>
<tr><td>Alpha 5</td><td>Learning, memory</td><td>Moderate</td><td>Low</td><td>Very Low</td></tr>
</table>

DUAL OREXIN ANTAGONISTS (DORA)

- Lemborexant (Dayvigo®), daridorexant (Quviviq®)
- New and novel drug class: orexins are neuropeptides that promote wakefulness when orexin-1 and orexin-2 receptors are activated
- MOA: DORAs inhibit orexin-1 and orexin-2 receptors → normalize sleep-wake function by reducing wakefulness and unwanted transitions between wake and sleep
- Low potential for misuse relative to benzodiazepines and Z-drugs
- Lemborexant and daridorexant are indicated for insomnia related to sleep onset and/or sleep maintenance
  - Wait at least 7 hours before driving or operating machinery
  - Sleep onset time may be delayed if taken with a high-fat meal
  - No evidence of withdrawal symptoms or rebound insomnia upon withdrawal
  - Has efficacy data for up to 12 months → tolerance not yet reported
  - General AEs: abnormal dreams, drowsiness, fatigue, headache, nightmares, complex sleep behaviours
  - Avoid combination with CNS depressants (e.g. alcohol, sedative hypnotics, opioids, TCA)
- Special populations:
  - Pregnancy/Breastfeeding: Lemborexant: only if benefits>risks, Daridorexant: no information available
  - Pediatric (< 18 years old): Not authorized by Health Canada, no safety/efficacy data

ANTIDEPRESSANTS

Reserve antidepressants for when other treatments fail or when associated with a co-morbid condition (e.g. depression, substance use disorder)

Doxepin (tricyclic antidepressant)

- MOA: acts as a selective histamine H1-receptor antagonist at very low doses (3–6 mg). The sedative effect is by antagonism of the histaminic part of the wake system, instead of promoting sleep by activating the GABAergic system
- ≤ 6 mg doses are officially indicated for insomnia, whereas significantly elevated doses are used for depression
- Improves sleep maintenance, taken 30 minutes before bedtime
- If substance abuse or dependence is a concern due to minimal physical tolerance/ dependence
- Has minimal next-day side effects and does not have typical TCA side effects at low doses
- Do not take within 3 hours of a meal due to delayed absorption and potential for next day drowsiness

Trazodone (5-HT_{2A}/H_1 antagonist)

- No official indication for insomnia, limited evidence for use
- Sometimes prescribed off-label for sleep onset or maintenance at low doses (e.g. 50-100 mg, though dose is patient specific)
- Low anticholinergic activity and short half-life provide a low risk of morning hangover effects
- Minimal risk of tolerance/dependence
- AEs: orthostatic hypotension, priapism (rare), cardiac conduction (rare)

SPECIAL POPULATIONS - OLDER ADULTS

- First-line: CBT-I is more effective than medication in older adults
- Individualize pharmacological treatment options based on patient-specific factors e.g. drug interactions, side effects
- Safe choices:
  - Doxepin (≤ 6 mg): safest and best studied in older adults for sleep maintenance issues
  - Melatonin: sleep-onset insomnia
  - Lemborexant
- Caution:
  - Benzodiazepines and Z-drugs → increased risk of falls, confusion and increased potency
    - Zolpidem may be preferred due to its shorter half-life
    - Low-dose eszopiclone (2 mg) has also demonstrated short-term efficacy (up to 24 weeks) in the elderly with a low risk of falls
- Safety profile unknown:
  - Low-dose trazodone and mirtazapine (off-label)
    - Commonly conflicting evidence and potential for falls and fractures
    - Reserve for insomnia with comorbid depression/anxiety or more resistant cases

SPECIAL POPULATIONS - PREGNANCY AND BREASTFEEDING

- First-line: CBT-I + address contributing comorbid illnesses in pregnancy (e.g. GERD, restless legs syndrome)

- Optimize non-pharmacological options e.g. sleep hygiene
- If severe + disabling → pharmacologic option weighing risk-benefit ratio
- Avoid hypnotic therapy when possible, as these may cause adverse effects in pregnancy (e.g. low birth weight in infants, preterm delivery)
- Melatonin lacks evidence for use in pregnancy

- Most drugs are excreted in breast milk
- Limited data, but Z-drugs appear relatively safe in breastfeeding with proper monitoring

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Therapeutic Goal	Timeframe for follow-up
# of hours of sleep per night	Normalize to ~ 7-8 hours (may vary based on patient's baseline)	Patient to monitor daily while on therapy. HCP to follow up within a few days to 1 week after therapy initiation
Time taken to fall asleep	Reduce to 30 minutes or less	
# of nighttime awakenings	Reduce to minimal or none	
Perceived quality of sleep	Improved	
Perceived daytime functioning		

MONITORING PARAMETERS

SAFETY ENDPOINTS		
Parameter	**Degree of Change**	**Timeframe**
Falls	Prevent	Duration of therapy
Memory impairment		
Morning grogginess	Prevent/minimize to tolerable levels	
Dizziness		
Abnormal dreams/nightmares		

TAKEAWAY

- First-line for chronic insomnia (all patients): CBT-I with or without pharmacotherapy
- Pharmacologic therapy at the lowest dose for the shortest term, episodic insomnia unresponsive to lifestyle changes and sleep hygiene → Z drugs and benzodiazepines
  - Taper down dose and frequency upon discontinuation of hypnotic medication to avoid rebound insomnia
- Avoid CNS depressant use with insomnia pharmacotherapy
- Health Canada-approved medications for insomnia include: DORAs (lemborexant, daridorexant), Z-Drugs (zopiclone, eszopiclone, zolpidem), benzodiazepines (flurazepam, temazepam, nitrazepam, triazolam), and doxepin
- Low-dose doxepin (3-6 mg) is the only antidepressant approved for insomnia
- Melatonin and valerian have insufficient data on efficacy
- If antihistamines are used, their application should be limited to a short duration

CASE 1

IN is a 55-year-old male who has been experiencing insomnia approximately 4 nights per week for the last 4 months. He has trouble falling asleep and tosses in bed all night. He is very concerned, as this has significantly affected his quality of life. IN has been stressed lately and thinks this is why he has not been sleeping. He feels lethargic at work due to lack of sleep and is concerned about his mental health at this point. IN has been diagnosed with Attention-Deficit/Hyperactivity Disorder (ADHD) for which he takes Biphentin® (methylphenidate controlled-release) 20 mg with dinner. He does not smoke or drink alcohol and has no known allergies to medications.

1. Which of the following is NOT considered a common medication cause of insomnia?
    a) Methylphenidate
    b) Metformin
    c) Caffeine pills
    d) Corticosteroids

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2. Which of the following next steps would be most appropriate?
   a) Decrease his dose of Biphentin®
   b) Change the administration time of Biphentin® to the morning
   c) A 7-day trial of a benzodiazepine
   d) Avoid exercise immediately before bedtime

CASE 1

Six months later, IN returns to your pharmacy. He tells you that he followed your advice and switched his Biphentin® administration time to the morning, with minimal improvement in his symptoms. IN also tried a combination of sleep hygiene techniques and melatonin. Unfortunately, these did not provide satisfactory improvement in his symptoms either. IN feels quite discouraged and finds that his insomnia significantly interferes with his daytime activities. He wonders if you have any further recommendations.

3. Which of the following prescription medications would be most appropriate for short-term use?
   a) Nitrazepam
   b) Temazepam
   c) Triazolam
   d) Flurazepam

CASE 1

IN completed a 7-day trial of the appropriate benzodiazepine but did not tolerate the significant hangover effects and still felt quite lethargic at work. IN mentions that he goes to bed around 10 PM and wakes up at 6 AM for work.

4. Which drug is the most appropriate alternative to try next?
   a) Doxepin 300 mg nightly
   b) Zopiclone 7.5 mg nightly
   c) Trazodone 300 mg nightly
   d) Lemborexant 5 mg nightly

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CHANGE LOG

July 2025
- Slide 13: Changed clonazepam from intermediate-acting to long-acting (although it falls somewhere in the middle)

Aug 2025
- Major formatting, wording/phrasing changes made throughout
- Updated legend
- Slide 5: Updated information
- Consolidated non-pharmacological measures into 1 slide
- Slide 12: Modified and added information
- Expanded Z-drugs to 2 slides
- Slide 16: Added information on food restrictions and special populations
- References slide updated

September 2025
- Slide 19-20: Added monitoring parameters

November 2025
- Rephasing edits, significant formatting changes made and slides re-arranged for better flow
- Slide 5: Simplified for clarity
- Slide 7: Minor update to goals of therapy
- Slide 9: Added slide on management of acute insomnia
- Slide 11: Added to clearly outline Health Canada-Approved agents
- Slide 12: Added information to third column
- Slide 18: Added that low doses of doxepin are indicated for insomnia
- Slides 24-27: Some changes to case and questions

March 2026
- Slide 2: Updated NAPRA competencies
- Slide 13: Simplified slide
- Slide 15: Minor updates
- Slide 20: Added information on breastfeeding